

Undergraduate Curriculum in Biological Systems Engineering Open Option

2023-2024 Catalog

Total Credits 128

First Year (32 cr.)

Semester 1 (16 cr.)			Semester 2 (16 cr.)		
R	Engr 101	Orientation (FS)	1	A B E 110	Experiencing BSE (S)
3	A B E 170	Graphics and Design (FS)	3	A B E 160	Engineering Problems (S)
4	Chem 167	General Chemistry (FS)	4	Math 166	Calculus II (FSSS)
1	Chem 167L	General Chemistry Lab (FS)	4	Phys 231	Classical Physics I (FSSS)
4	Math 165	Calculus I (FSSS)	1	Phys 231L	Classical Physics I Lab (FS)
3	Engl 150	Crit. Think. and Comm.(FSSS)	3	Engl 250	Writ/Oral/Vis/Elec Composition (FSSS)
1	Lib 160	Information Literacy (FSSS)			

Second Year (31 cr.)

Semester 3 (16 cr.)			Semester 4 (15 cr.)		
3	A B E 216	Fund. Ag. and Biol. Engineering (F)	2	A B E 218	Project Mgmt/Design Ag & Bio Engr (S)
3	C E 274	Statics of Engineering (FSSS)	1	A B E 201	Preparing for the Workplace (FS)
3	Biol 212	Principles of Biology II (FS)	1	A B E 273	2D Auto CAD (FS)
3	Chem 331	Organic Chemistry I (FSSS)	3	M E 231	Thermodynamics (FSSS)
1	Chem 331L	Organic Chemistry I Lab (FSSS)	3	Chem 332	Organic Chemistry II (FSSS)
3	→	US Diversity Elective (FSSS)	1	Chem 332L	Organic Chemistry II Lab (FSSS)
			4	Math 267	Differential Equations (FSSS)

Third Year (32 cr.)

Semester 5 (17 cr.)			Semester 6 (16 cr.)		
3	A B E 316	Applications and Systems Modeling (FS)	4	A B E 363	Electric Power and Electronics (FS)
3	A B E 378	Mechanics of Fluids (FS)	3	A B E 380	Principles of BSE (S)
3	Micro 302	Biology of Microorganisms (FSSS)	3	E M 324	Mechanics of Materials (FSSS)
1	Micro 302L	Microbiology lab (FSSS)	3	I E 305	Engineering Economic Analysis (FSSS)
3	STAT 305	Engineering Statistics (FSSS)	3	→	Sequence Course I (FS)
3	→	Communication Elective (FSSS)			
1	→	Lab Elective (ABE 378L preferred)			

Fourth Year (33 cr.)

Semester 7 (17cr.)			Semester 8 (15 cr.)		
2	A B E 415	Ag & Biosystems Engr. Design I (FS)	2	A B E 416	Ag & Biosystems Engr. Design II (FS)
3	A B E 404	Instrumentation for Ag and Bio Engr (F)	3	A B E 451	Food and Bioprocess Engineering (S)
3	A B E 480	Engineering Analysis of Bio Systems (F)	4	M E 436	Heat Transfer (FSSS)
3	→	Sequence Course II (FS)	3	→	Sequence Course III (FS)
3	→	International Perspectives Elect. (FSSS)	3	→	SS/H Elective (FSSS)
3	→	SS/H Elective (FSSS)			

**Please check the current catalog and Schedule of Classes for most recent offerings*